

3R Robotic Arm

Project Report by

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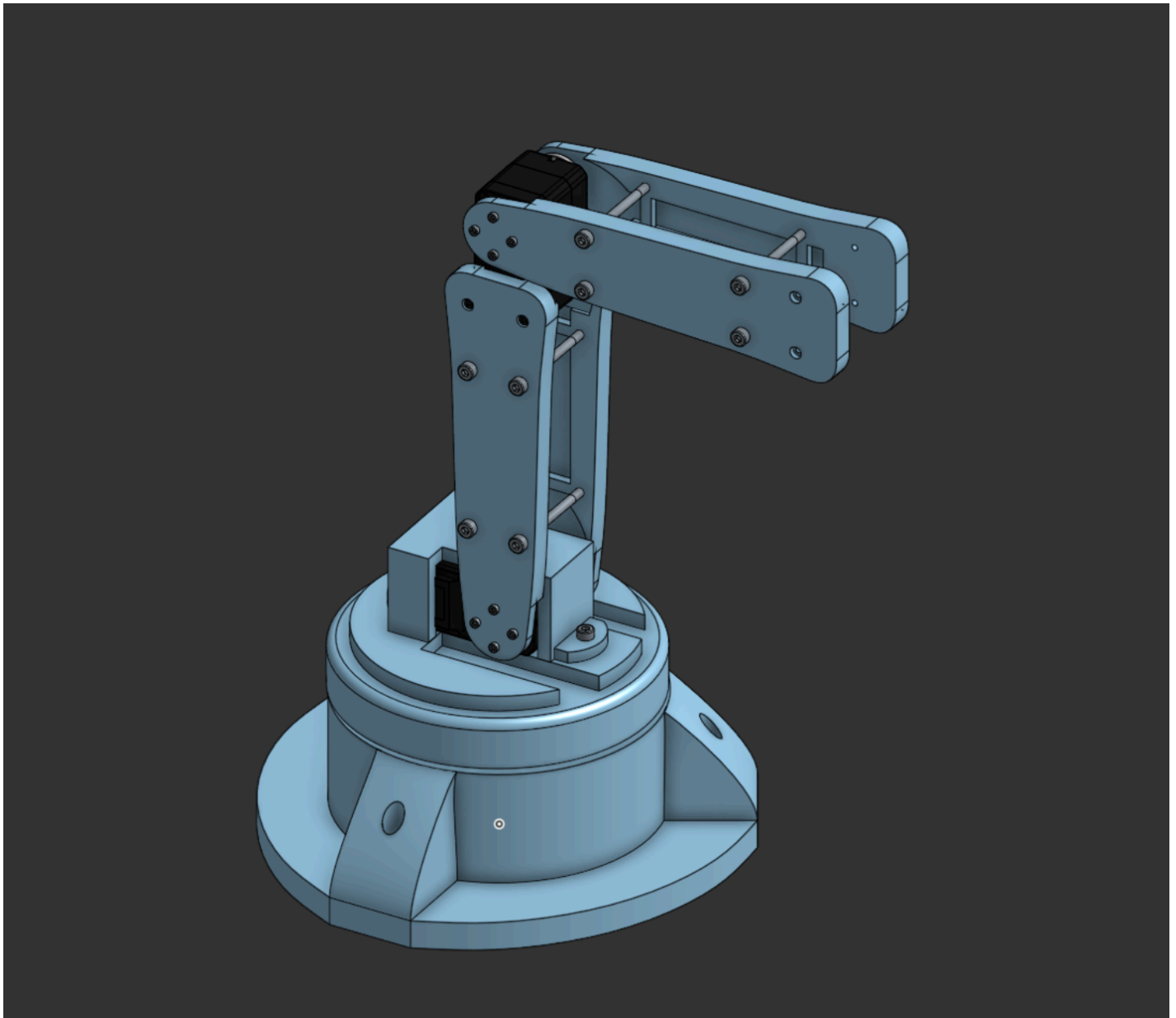
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Introduction

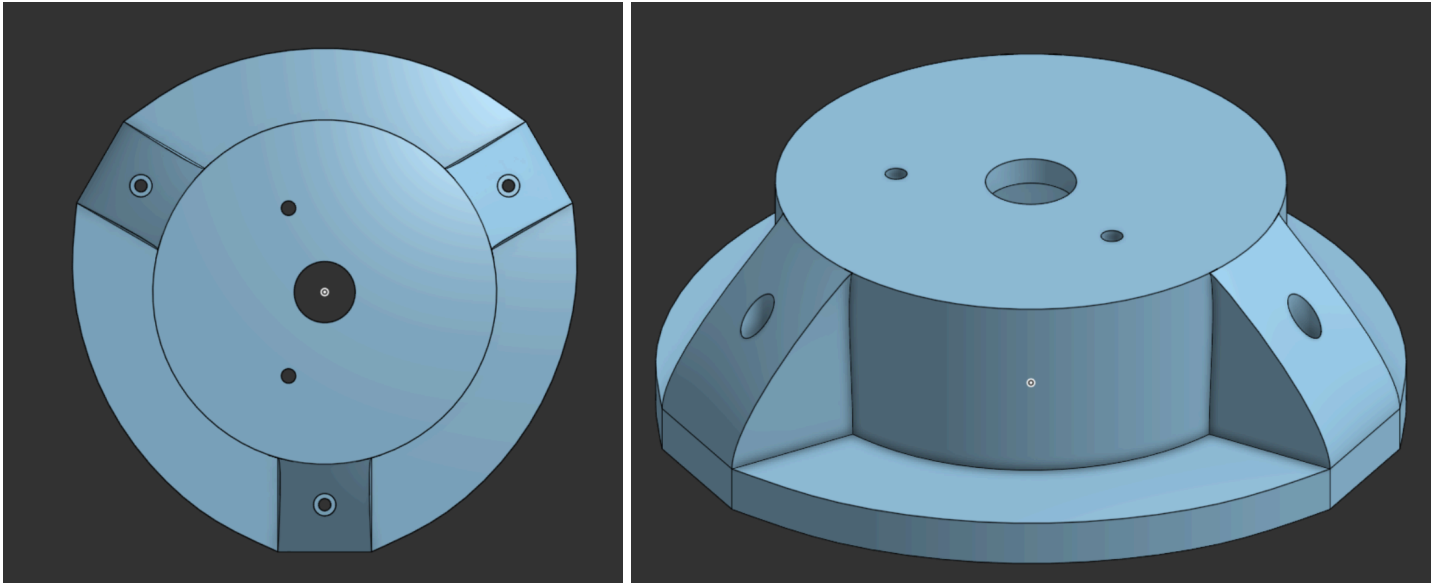
The Robotic Arm is a standard robotic system that acts as an introduction to test the foundations of sequential assembly, actuation, transformations, and moving frames. The system that had been modeled is a 3R (R for revolute) jointed robotic arm, where one revolute is for yaw of the base and the two subsequent links that articulate within a single plane. For an open kinematic chain, the number of joints determines its degrees of freedom (DOF); hence, for this system, the degrees of freedom are three. Typically, modeling a 3R arm is easy if it is within the scope of simulation, as it can be simplified; however, when considering the scope of physical hardware, the sequence of assembly is indispensable. The general thought process to approach this will be:

1. Fixing the base to a stable surface
2. Assembly of the turntable base
3. Assembly of the motor's housing
4. Assembly of individual links

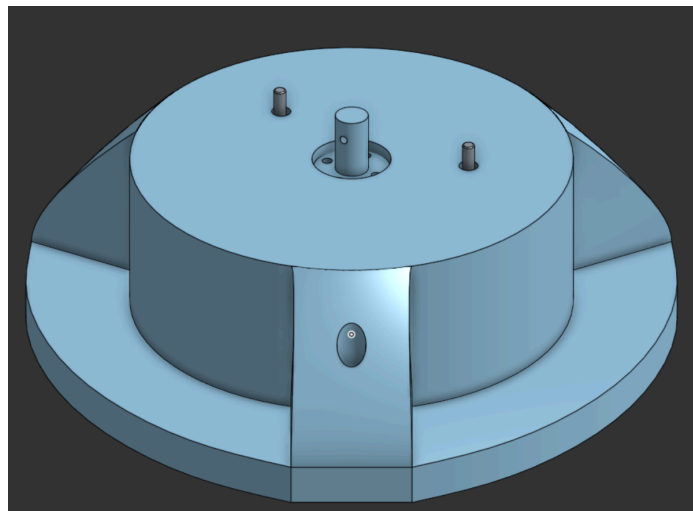
Additionally, we also need to consider the tolerances and manufacturing methods that are apt for the individual components.



Fixing the base to a stable surface:

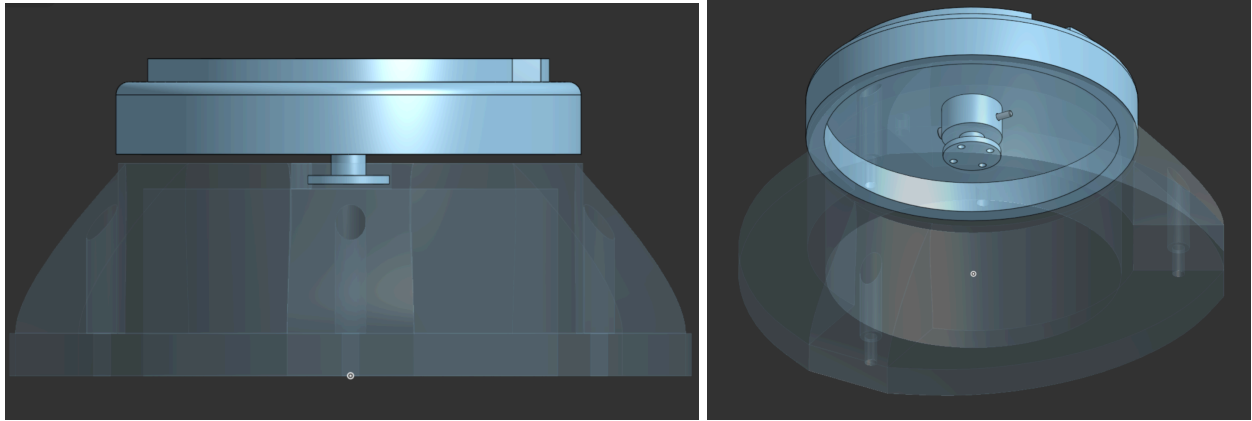


Having an anchoring support for the robotic arm is essential for its stability. As the bot is of smaller dimensions, three anchored locations are more than enough to stabilize the bot. The three anchors are actually screws that are screwed into the base and then attached to a firm wooden plank. An extra wing is added on the sides of the base to distribute the stress, preventing it from acting just on those two anchors at a time while lifting. The central hole is for the flange shaft from the motor, which drives the turntable. The two minor holes are for screws that secure the motor housing.

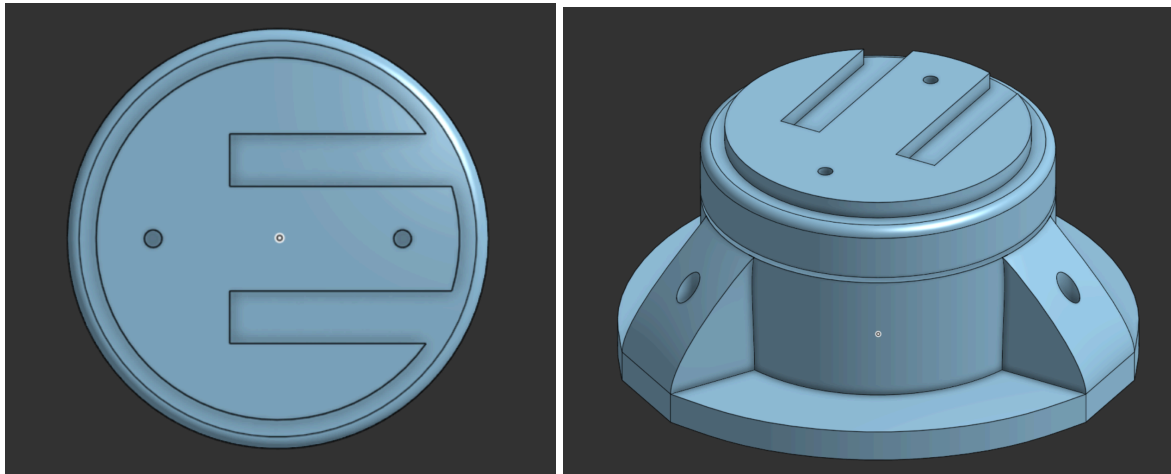


The motors used are ST3215, and they come with a flange shaft for attachment, which is made with stainless steel. If that shaft was made with 3D printing using PLA, it is weak along its layers; hence, torsion can cause it to break. Therefore, the flange shaft is made of stainless steel, which is preferred.

Assembly of the turntable base:

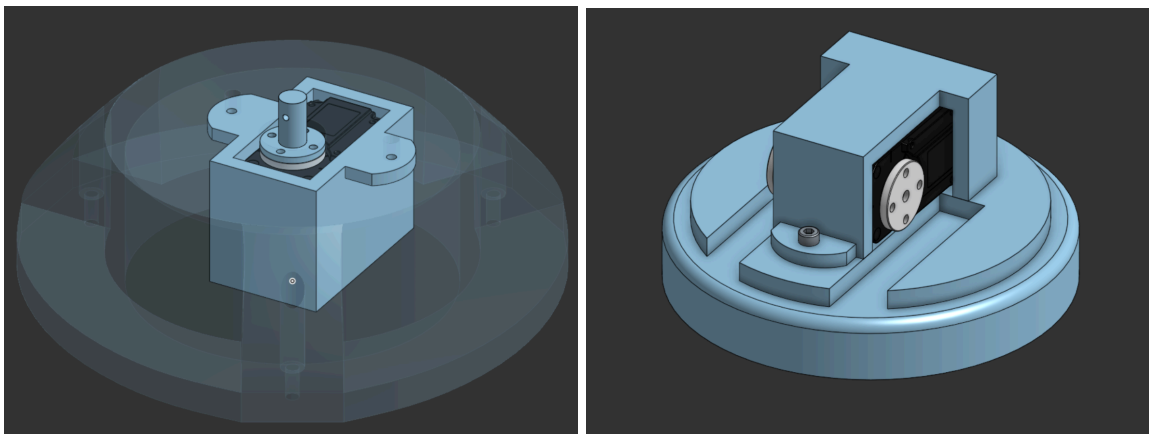


The turntable base has the sole goal, as the name implies, to turn with respect to the fixed base. The flange shaft has a hole through which a screw can be inserted as a lock. In the turntable piece, a similar pair of holes is created to secure the flange shaft to the turntable. On top of the turntable, some leeway is made to prevent the joints of the second motor and link from colliding with the turntable. The two minor holes are left for heated inserts to attach the housing of the motor firmly.

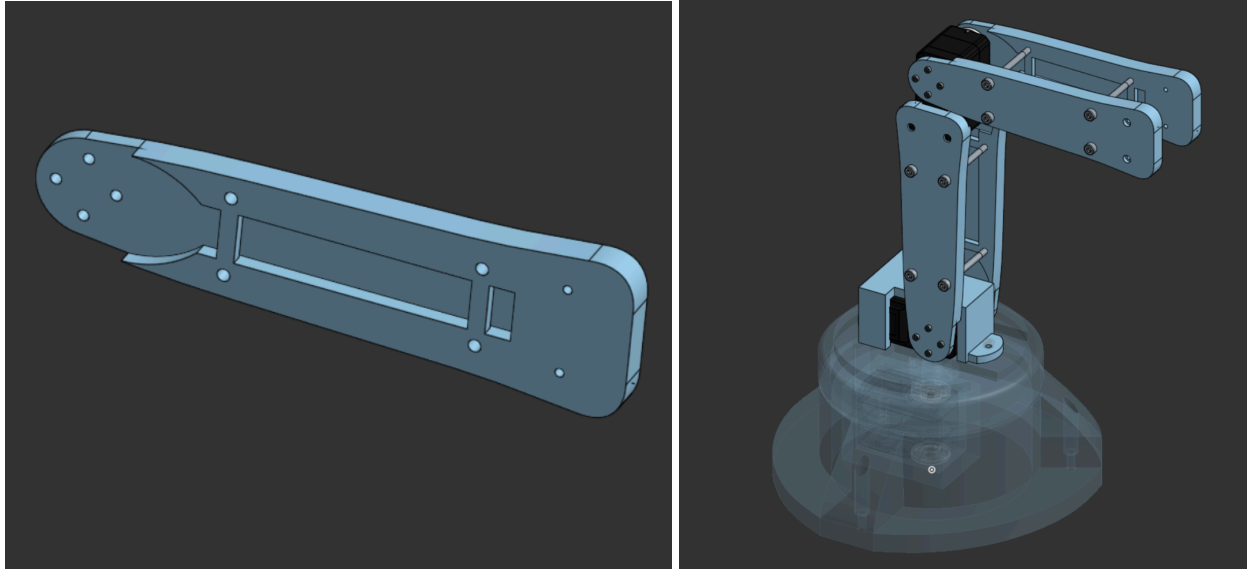


Assembly/housing of motor:

The first image is the housing for the motor in the fixed base, and the second image is the motor housing on the turntable base. Both the housing encloses the motors firmly to prevent relative movement between them. This ensures proper actuation and reduced error.

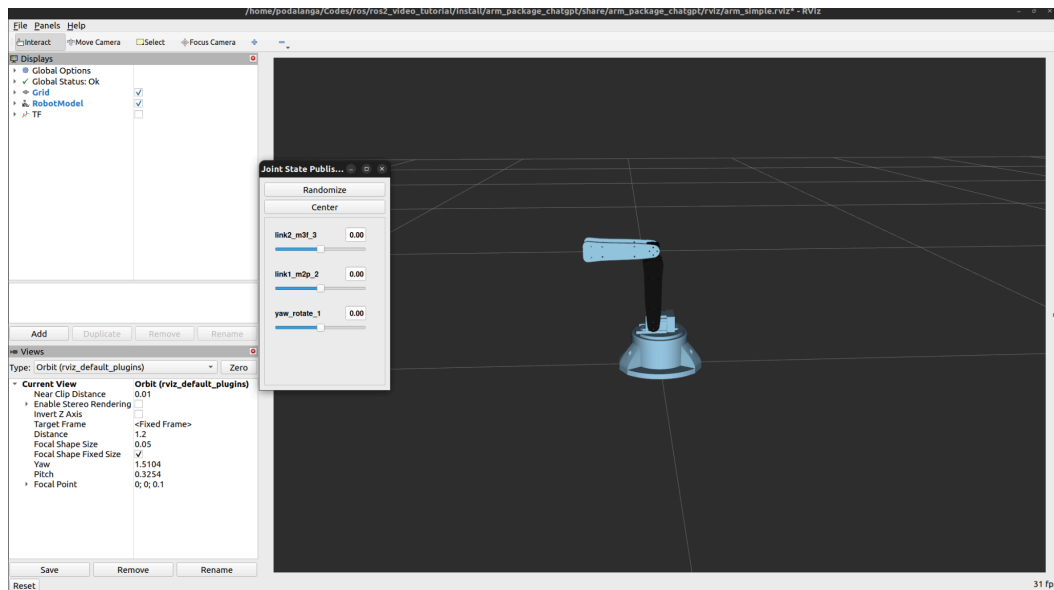


Assembly of individual links:



From the second motor, two links are screwed firmly to ensure its accurate first planar revolute movement. At the end of the first two links, the third motor is attached, which is for the second planar revolute movement. Consider the first two links from the second motor to the third motor as “first pair of links” and the pair of links from the third motor to the TCP as the “second pair of links”. Between each pair of links, allen screws are used as rigid supports. 3D printing a support can easily break, and its strength is highly dependent on the layering; hence, it is not recommended.

URDF and Visualization:



URDF of the arm was created using the onshape-to-robot extension and was actuated in RVIZ2.

Attachments:

CAD model [link](#)

Hardware assembly [link](#)

RVIZ2 Simulation [link](#)